Schrödinger Bridge / Dynamic Optimal Transport in 1D

Problem Setup, Discretization, and ADMM Algorithm

March 5, 2026

Contents

1 Problem Setup **2**
- 1.1 Continuous Formulation (Benamôu–Brenier) . . . 2
- 1.2 Regularization (Schrödinger Bridge) . . . 2

2 Staggered Grid Discretization **2**
- 2.1 Grid Parameters . . . 2
- 2.2 Variable Locations . . . 2
- 2.3 Space-Time Grid Diagram . . . 3
- 2.4 Finite Difference Operators . . . 3
- 2.5 Interpolation Operators . . . 3
- 2.6 Fast Poisson Solve via DCT . . . 3

3 ADMM Algorithm **4**
- 3.1 Splitting Formulation . . . 4
- 3.2 Iteration . . . 4
- 3.3 Proximal Operator Detail . . . 5
- 3.4 Projection Step Detail . . . 5
- 3.5 Convergence Criterion . . . 6

4 Implementation Summary **6**
- 4.1 Code Structure . . . 6
- 4.2 Solver Interface . . . 6
- 4.3 Initialization . . . 6
- 4.4 Parameter Guidance . . . 6

5 Numerical Results (Gaussian Translation) **7**

1 Problem Setup

1.1 Continuous Formulation (Benamôu–Brenier)

We seek a curve of probability densities $\rho(t,x)$ and a momentum field $m(t,x)$ on the space-time domain $[0,1]\times[0,1]$ solving the *dynamic optimal transport* problem:

$$\min_{\rho,m} \int_0^1 \int_0^1 \frac{m(t,x)^2}{2\,\rho(t,x)}\,dx\,dt \tag{1}$$

subject to the **continuity equation**

$$\partial_t \rho + \partial_x m = 0, \tag{2}$$

$$\rho(0,\cdot) = \rho_0, \quad \rho(1,\cdot) = \rho_1, \tag{3}$$

$$m(t,0) = m(t,1) = 0, \tag{4}$$

$$\rho \geq 0. \tag{5}$$

The integrand $m^2/(2\rho)$ is the *kinetic energy density*. At the optimum the objective equals $W_2(\rho_0,\rho_1)^2$, the squared 2-Wasserstein distance.

1.2 Regularization (Schrödinger Bridge)

A diffusion coefficient $\varepsilon \geq 0$ replaces the continuity constraint with the *Fokker–Planck equation*:

$$\partial_t \rho + \partial_x m = \varepsilon\,\Delta_x \rho. \tag{6}$$

As $\varepsilon \to 0$ the problem reduces to unregularized W_2 transport. Positive ε connects to the **Schrödinger bridge** problem and smooths the solution, aiding numerics.

2 Staggered Grid Discretization

2.1 Grid Parameters

Symbol	Value	Meaning
N_x	user-set	number of spatial cells
N_t	user-set	number of temporal slabs
Δx	$1/N_x$	spatial cell width
Δt	$1/N_t$	temporal slab height

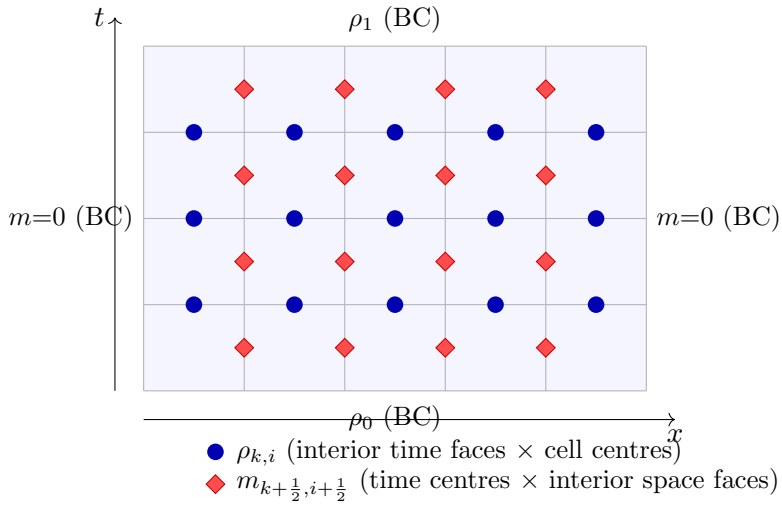
2.2 Variable Locations

Different unknowns live at *different grid locations* (MAC / staggered layout), which ensures exact discrete conservation:

Variable	Time location	Space location	Array size
ρ	interior time faces $t_k = k\Delta t$	cell centres	$(N_t-1)\times N_x$
m	time cell centres $t_{k+\frac{1}{2}}$	interior space faces	$N_t \times (N_x-1)$

Boundary data $\rho(0,\cdot)=\rho_0$ and $\rho(1,\cdot)=\rho_1$ are held fixed; the spatial boundary momenta $m(\cdot,0)=m(\cdot,1)=0$ are enforced by restricting to interior faces only.

2.3 Space-Time Grid Diagram



2.4 Finite Difference Operators

Four first-order stencils cover the two directions and two grid offsets. Below D_t^ϕ denotes the time derivative whose output sits at *time-face* locations (used in the continuity equation):

$$(D_t^\phi \rho)_k = \frac{\rho_k - \rho_{k-1}}{\Delta t}, \qquad k = 1, \ldots, N_t - 1, \tag{7}$$

with boundary values $\rho_0 = \rho_0$ and $\rho_{N_t} = \rho_1$. The transpose $D_t^\rho = -(D_t^\phi)^\top$ maps from faces to cell centres. Analogous operators D_x^ϕ and D_x^m act in the spatial direction.

The **discrete continuity equation** at each space-time cell (k, j) is:

$$\frac{\rho_{k+1,j} - \rho_{k,j}}{\Delta t} + \frac{m_{k+\frac{1}{2},j+\frac{1}{2}} - m_{k+\frac{1}{2},j-\frac{1}{2}}}{\Delta x} = 0. \tag{8}$$

2.5 Interpolation Operators

Because m^2/ρ couples variables at different locations, we define second-order averaging operators:

$$(I_t^\phi \rho)_k = \tfrac{1}{2}(\rho_{k-\frac{1}{2}} + \rho_{k+\frac{1}{2}}), \qquad (I_x^\phi m)_j = \tfrac{1}{2}(m_{j-\frac{1}{2}} + m_{j+\frac{1}{2}}), \tag{9}$$

bringing both fields to the space-time **cell centres** where the kinetic energy is evaluated.

Adjoint note: When the forward operator is affine, $L(u) = Au + b$ (with b encoding BCs), the adjoint is $L^*(v) = A^\top v$ — the BC term is *not* included. Using non-zero BCs in the adjoint corrupts the gradient direction and leads to incorrect solutions.

2.6 Fast Poisson Solve via DCT

The projection step requires inverting the operator $(-\Delta_{xt} + \varepsilon^2 \Delta_x^2)$ with Neumann BCs, where $\Delta_{xt} = \partial_{tt} + \Delta_x$ is the space-time Laplacian and Δ_x^2 is the spatial biharmonic. Because the

discrete Neumann Laplacian diagonalises in the DCT basis, the eigenvalues are precomputed once:

$$\lambda_x^\ell = \frac{2 - 2\cos(\pi \Delta x\, \ell)}{\Delta x^2}, \quad \ell = 0, \ldots, N_x - 1, \tag{10}$$

$$\lambda_t^k = \frac{2 - 2\cos(\pi \Delta t\, k)}{\Delta t^2}, \quad k = 0, \ldots, N_t - 1, \tag{11}$$

$$\lambda^{k,\ell} = \lambda_t^k + \lambda_x^\ell + \varepsilon^2 (\lambda_x^\ell)^2. \tag{12}$$

The solve $\hat{\phi} = \widehat{\mathrm{RHS}} \,/\, \lambda$ is $O(N_x N_t \log N_x N_t)$, and the zero-mode $\lambda^{0,0} = 0$ is handled by setting $\hat{\phi}^{0,0} = 0$ (fixing the additive gauge of ϕ).

3 ADMM Algorithm

Optimize-first philosophy. ADMM is applied directly to the *continuous* (infinite-dimensional) Benamou–Brenier problem (1)–(6). The three ADMM steps — proximal map, projection, and dual update — are derived in the continuous function-space setting. Only afterwards is each step independently discretized onto the staggered grid of Section 2. This *optimize-first, discretize-second* ordering means the discrete iteration does not fit the standard finite-dimensional ADMM framework, and we currently have no formal convergence guarantee; convergence is monitored empirically via the residual (22).

3.1 Splitting Formulation

We introduce an auxiliary copy $(\tilde{\rho}, \bar{m})$ of the primal variables and rewrite (1)–(6) as the split problem

$$\min_{\rho, m, \tilde{\rho}, \bar{m}} \underbrace{\int_0^1 \int_0^1 \frac{m^2}{2\,\rho}\, dx\, dt}_{f(\rho,m)} + \underbrace{\iota_{\mathcal{C}}(\tilde{\rho}, \bar{m})}_{g(\tilde{\rho},\bar{m})} \quad \text{s.t.} \quad (\rho, m) = (\tilde{\rho}, \bar{m}), \tag{13}$$

where $\mathcal{C}$ is the *continuous* Fokker–Planck constraint set:

$$\mathcal{C} = \big\{(\rho, m) : \partial_t \rho + \partial_x m = \varepsilon\, \Delta_x \rho,\ \rho(0,\cdot) = \rho_0,\ \rho(1,\cdot) = \rho_1,\ m(\cdot, 0) = m(\cdot, 1) = 0\big\}.$$

Each ADMM step below is derived for this continuous problem and then discretized independently.

3.2 Iteration

With penalty parameter $\gamma > 0$ and dual variables (δ_ρ, δ_m), the three steps per iteration are:

Step 1 – Proximal step (kinetic energy).

$$(\rho^{n+1}, m^{n+1}) \leftarrow \mathrm{prox}_{f/\gamma}\Big(\tilde{\rho}^n + \tfrac{\delta_\rho^n}{\gamma},\ \bar{m}^n + \tfrac{\delta_m^n}{\gamma}\Big). \tag{14}$$

Step 2 – Projection step (continuity constraint).

$$(\tilde{\rho}^{n+1}, \bar{m}^{n+1}) \leftarrow \mathrm{proj}_{\mathcal{C}}\Big(\rho^{n+1} - \tfrac{\delta_\rho^n}{\gamma},\ m^{n+1} - \tfrac{\delta_m^n}{\gamma}\Big). \tag{15}$$

Step 3 – Dual update.

$$\delta_\rho^{n+1} \leftarrow \delta_\rho^n - \gamma\big(\rho^{n+1} - \tilde{\rho}^{n+1}\big), \tag{16}$$

$$\delta_m^{n+1} \leftarrow \delta_m^n - \gamma\big(m^{n+1} - \bar{m}^{n+1}\big). \tag{17}$$

3.3 Proximal Operator Detail

The kinetic energy $|m|^2/\rho$ is defined *pointwise*, so the continuous-level proximal operator acts cell-by-cell at space-time cell centres (the phi-grid). Because ρ and m live on staggered grids, the implementation requires three substeps:

1. **Interpolate inputs to cell centres.**

$$\bar{\rho} = I_t^\phi\Big(\tilde{\rho}^n + \tfrac{\delta_\rho^n}{\gamma}\Big), \qquad \bar{m} = I_x^\phi\Big(\bar{m}^n + \tfrac{\delta_m^n}{\gamma}\Big).$$

   The boundary conditions (ρ_0, ρ_1, and zero flux) are folded into the interpolation operators I_t^ϕ and I_x^ϕ.

2. **Pointwise prox at cell centres.** At each cell centre, given inputs $(\tilde{r}, \tilde{\mu}) = (\bar{\rho}, \bar{m})$ and $\sigma = 1/\gamma$, solve

$$\min_{r,\mu} \frac{\mu^2}{2r} + \frac{\gamma}{2}\big[(r-\tilde{r})^2 + (\mu-\tilde{\mu})^2\big]. \tag{18}$$

   The optimal momentum is $\mu = r\tilde{\mu}/(r+\sigma)$, and r satisfies the cubic

$$r^3 + (2\sigma - \tilde{r})\, r^2 + (\sigma^2 - 2\sigma\tilde{r})\, r - \sigma\Big(\sigma\tilde{r} + \tfrac{\tilde{\mu}^2}{2}\Big) = 0. \tag{19}$$

   The unique positive real root is found via Cardano's method (`solve_cubic.m`). Cells with $r \le 10^{-12}$ are clamped to zero (vacuum truncation).

3. **Interpolate outputs back to staggered grids.**

$$\rho^{n+1} = I_t^\rho(r), \qquad m^{n+1} = I_x^m(\mu).$$

This three-step procedure is a direct discretization of the continuous proximal map; there is no guarantee that the result equals the exact prox of the discrete kinetic-energy functional.

3.4 Projection Step Detail

Given (μ, ψ), find $(\rho_{\text{out}}, m_{\text{out}})$ in $\mathcal{C}$ by introducing a correction potential ϕ:

$$\rho_{\text{out}} = \mu + \partial_t\phi + \varepsilon\,\Delta_x\phi, \qquad m_{\text{out}} = \psi + \partial_x\phi. \tag{20}$$

Substituting into the continuous Fokker–Planck equation (6) yields the elliptic PDE

$$\underbrace{(-\Delta_{xt} + \varepsilon^2\Delta_x^2)}_{\text{fast solve via DCT}}\phi = \partial_t\mu + \partial_x\psi - \varepsilon\,\Delta_x\mu, \tag{21}$$

solved in $O(N_x N_t \log N_x N_t)$ via (10)–(12). This step is then discretized using the staggered operators of Section 2.

3.5 Convergence Criterion

We monitor the change in the projected variable:

$$\mathrm{res}^n = \sqrt{\Delta x\, \Delta t \left(\left\| \tilde{\rho}^{n+1} - \tilde{\rho}^n \right\|^2 + \left\| \bar{m}^{n+1} - \bar{m}^n \right\|^2 \right)}. \tag{22}$$

Convergence requires $\mathrm{res}^n < \mathrm{tol}$.

For robustness of the extra-gradient variant, the ADMM step size must satisfy

$$\frac{\tau}{\gamma} > \|A^\top A\|_2 \approx 1, \tag{23}$$

where A is the staggered interpolation operator. We recommend a ratio $\tau/\gamma \approx 1.5$ (50 % margin).

4 Implementation Summary

4.1 Code Structure

File	Purpose
`sb1d_admm.m`	Main solver: ADMM loop + all helper functions
`solve_cubic.m`	Vectorised Cardano cubic root finder
`mirt_dctn.m`	N-D Discrete Cosine Transform
`mirt_idctn.m`	N-D Inverse DCT
`test_adjoints.m`	Numerical adjoint consistency checks
`test_operators.m`	Staggered operator sanity tests

4.2 Solver Interface

```
[rho, mx, outs] = sb1d_admm(rho0, rho1, opts)

opts.nt      = 64;      % temporal slabs
opts.gamma   = 10;      % ADMM penalty
opts.maxIter = 2000;    % max iterations
opts.vareps  = 1e-4;    % diffusion regularisation (0 = pure OT)
```

4.3 Initialization

The density is initialised by linear interpolation: $\rho^0(t,\cdot) = (1-t)\rho_0 + t\rho_1$, with $m^0 = 0$. This satisfies the continuity equation trivially and provides a natural warm start.

4.4 Parameter Guidance

Parameter	Typical range	Effect
γ	1–100	ADMM step size; larger $\Rightarrow$ faster but less stable
ε	10^{-4}–10^{-2}	Diffusion; smaller $\Rightarrow$ closer to W_2 OT
N_x, N_t	64–256	Spatial/temporal resolution

5 Numerical Results (Gaussian Translation)

Test case: $\rho_0 = \mathcal{N}(1/3, 0.05^2)$, $\rho_1 = \mathcal{N}(2/3, 0.05^2)$ on $[0, 1]$. The exact W_2 geodesic translates the Gaussian mean linearly in t, with constant velocity $v(t, x) = 1/3$.

Expected behaviour:

- Density $\rho(t, \cdot)$ is Gaussian with mean $\frac{1}{3} + \frac{1}{3}t$ for all t.
- Velocity field $v = m/\rho$ is spatially uniform and time-constant.
- Residual (22) decreases monotonically to machine precision.

Grid: $N_x = 128$, $N_t = 64$, $\gamma = 10$, $\varepsilon = 10^{-4}$, 2000 iterations.